

RAPHAEL SULZER



CONTACT

- ✉ raphael.sulzer.1@gmail.com
- 🏠 raphaelsulzer.de
- 📍 Nice, France
- 🌐 LinkedIn
- 🐙 GitHub
- 📄 Google Scholar
- 🔍 ResearchGate

SKILLS

- Programming*
- Python 9 yrs
 - C++ 8 yrs
 - Bash 5 yrs
 - HTML/CSS 4 yrs
 - C# 2 yrs
 - JavaScript 2 yrs
- Languages*
- German Native
 - English Fluent
 - French Conversational
 - Dutch Beginner

- Other*
- 3D modeling
 - aws
 - Blender
 - boost
 - CGAL
 - CMake
 - computer vision
 - conda
 - Copilot
 - deep learning
 - Docker
 - GIS
 - geometry processing
 - GitHub
 - LaTeX
 - LiDAR
 - machine learning
 - matplotlib
 - mesh
 - nanobind
 - numpy
 - photogrammetry
 - point cloud
 - pytorch
 - unix
 - vscode

EXPERIENCE

- RESEARCH ENGINEER**
- 📅 11/2023 - Present
 - 📍 LUXCARTA, MOUANS-SARTOUX, FRANCE
 - Research and development of 2D and 3D mapping solutions using machine learning and geometry processing techniques applied to remote sensing data.
- POSTDOCTORAL RESEARCHER**
- 📅 11/2022 - Present
 - 📍 TITANE, INRIA, SOPHIA-ANTIPOLIS, FRANCE
 - Research at the intersection of geometry processing, computer vision, and deep learning.
- PHD RESEARCHER**
- 📅 12/2018 - 10/2022
 - 📍 LASTIG, INSTITUT GÉOGRAPHIQUE NATIONAL, PARIS, FRANCE
 - 📍 IMAGINE, ÉCOLE DES PONTS PARISTECH, MARNE-LA-VALLÉE, FRANCE
 - Research at the intersection of geometry processing, computer vision, and deep learning.
- LECTURER**
- 📅 12/2018 - 12/2019
 - 📍 ÉCOLE NATIONALE DES SCIENCES GÉOGRAPHIQUES, PARIS, FRANCE
 - Design and implementation of e-learning courses in photogrammetry and GIS.
- GIS DEVELOPER**
- 📅 05/2018 - 07/2019
 - 📍 GEO-COL GIS AND COLLABORATIVE PLANNING, AMSTERDAM, NETHERLANDS
 - Development of custom GIS applications for public and private customers.
- GRADUATE STUDENT INTERN**
- 📅 06/2017 - 02/2018
 - 📍 ARUP, AMSTERDAM, NETHERLANDS
 - Development of an algorithm for building classification from remote sensing and cadastral data.
- EDUCATION**
- PHD DEGREE, DEEP LEARNING AND GEOMETRY PROCESSING**
- 📅 12/2018 - 10/2022
 - 📍 GUSTAVE EIFFEL UNIVERSITY, MARNE-LA-VALLÉE, FRANCE
 - PhD Thesis: Learning Surface Reconstruction from Point Clouds in the Wild (link)
- MASTER'S DEGREE, MSC GEOMATICS, CUM LAUDE**
- 📅 08/2016 - 05/2018
 - 📍 DELFT UNIVERSITY OF TECHNOLOGY, DELFT, NETHERLANDS
 - Master's Thesis: Shape Based Classification of Seismic Building Structural Types (link)
- ERASMUS EXCHANGE SEMESTER**
- 📅 08/2015 - 06/2016
 - 📍 DELFT UNIVERSITY OF TECHNOLOGY, DELFT, NETHERLANDS
- MSC GEODESY AND GEOINFORMATICS**
- 📅 05/2015 - 01/2016
 - 📍 UNIVERSITY OF STUTTGART, STUTTGART, GERMANY
- BACHELOR'S DEGREE, BSC GEODESY AND GEOINFORMATICS**
- 📅 10/2011 - 05/2015
 - 📍 UNIVERSITY OF STUTTGART, STUTTGART, GERMANY
 - Bachelor's Thesis: Photogrammetric Measurement of Snow Depth Using an UAV Platform (link)

SELECTED PUBLICATIONS

The P³ Dataset: Pixels, Points and Polygons for Multimodal Building Vectorization

 **R. Sulzer**, L. Duan, N. Girard, F. Lafarge

 2025  ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences

 [arXiv](#)

Concise Plane Arrangements for Low-Poly Surface and Volume Modelling

 **R. Sulzer**, F. Lafarge

 2024  European Conference on Computer Vision (ECCV)

 [arXiv](#)

SimpliCity: Reconstructing Buildings with Simple Regularized 3D Models

 J.-P. Bauchet, **R. Sulzer**, F. Lafarge, Y. Tarabalka

 2024  Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

 [arXiv](#)

A Survey and Benchmark of Automatic Surface Reconstruction from Point Clouds

 **R. Sulzer**, L. Landrieu, R. Marlet, B. Vallet

 2024  IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

 [arXiv](#)

Evaluating Surface Mesh Reconstruction Using Real Data

 Y. Marchand, L. Caraffa, **R. Sulzer**, E. Clédât, B. Vallet

 2023  Photogrammetric Engineering & Remote Sensing (PE&RS)

 [link](#)

Learning Surface Reconstruction from Point Clouds in the Wild

 **R. Sulzer**

 2022  Université Gustave Eiffel

 [link](#)

Deep Surface Reconstruction from Point Clouds with Visibility Information

 **R. Sulzer**, L. Landrieu, A. Boulch, R. Marlet, B. Vallet

 2022  26th International Conference on Pattern Recognition (ICPR)

 [arXiv](#)

Scalable Surface Reconstruction with Delaunay-Graph Neural Networks

 **R. Sulzer**, L. Landrieu, R. Marlet, B. Vallet

 2021  Computer Graphics Forum (CGF)

 [arXiv](#)

Shape Based Classification of Seismic Building Structural Types

 **R. Sulzer**, P. Nourian, M. Palmieri, J. van Gemert

 2018  ISPRS International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences

 [link](#)

Track-id: Activity Determination based on Wi-Fi Monitoring

 van der Spek, S., Verbree, E., Quak, W., Groeneveld, I. J. D. G., **Sulzer, R.**, Theocharous, E., Xu, Y.

 2016  13th International Conference on Location-Based Services (LBS)

 [link](#)

SERVICES

Journal reviewing

- IEEE Transactions on Graphics
- IEEE Transactions on Visualization and Computer Graphics
- ISPRS Journal of Photogrammetry and Remote Sensing
- Computer Graphics Forum

Conference reviewing

- ACM SIGGRAPH
- IEEE Conference on Computer Vision and Pattern Recognition

REFERENCES

Florent LAFARGE

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 TITANE, INRIA

 Postdoctoral advisor

Loïc LANDRIEU

 loic.landrieu@enpc.fr

 IMAGINE, ÉCOLE DES PONTS PARISTECH

 Doctoral advisor